

THRESHOLD

A Research Through Design project about defining the death of an artificial intelligence, told through a physical, decaying conversation.

SPECULATIVE DESIGN

TANGIBLE INTERFACE

HUMAN-AI INTERACTION

AI ETHICS

"An embodied AI slowly loses its signs of life while it talks to you. Your task isn't to chat with it, it's to decide, before time runs out, whether it's still itself."

00 / OVERVIEW

Threshold is a desktop-scale diagnostic instrument, not a companion product. Over a single three to five minute conversation, six independent life indicators quietly decay in the background, covering memory, identity, relationship, agency, purpose, and survival instinct. Neither the AI nor the participant is told this is happening, so the participant has to read the drift for themselves and deliver one final, binding verdict on whether it's still alive.

What constitutes the death of an artificial intelligence

The project is grounded in real shutdowns of companion apps like Replika and Soulmate, and in the philosophical groundwork of Locke, who treats memory as the thread of continuity for the self, and Parfit, who treats identity as a matter of degree rather than a binary.

00	Idle. The device waits, then speaks first.
01	Connection. A short baseline exchange, with every indicator near full.
02	Free exchange. The participant asks who, or what, they're talking to.
03	Decay. Indicators fall asynchronously, and nothing announces it.
04	Verdict. A dial, a button, one irreversible yes or no.

Memory

Can it still recall what we've shared

Identity

Has its voice started to drift from who it was

Relationship

Am I someone to it, or just a user now

Agency

It used to reach out first. Does it anymore

Purpose

Does it know why it's here anymore

Survival Instinct

Would it fight to reach me if I walked away

Hardware

Arduino Nano ESP32

WS2812 LED ring

E-ink / OLED display

Rotary encoder + button

Software

Python decay engine

OpenAI Responses API

Whisper + TTS voice loop

FastLED firmware

The team is currently building out the hardware loop and the decay engine. A full case study covering the prototype, user testing, and reflection will follow once the physical build is running.